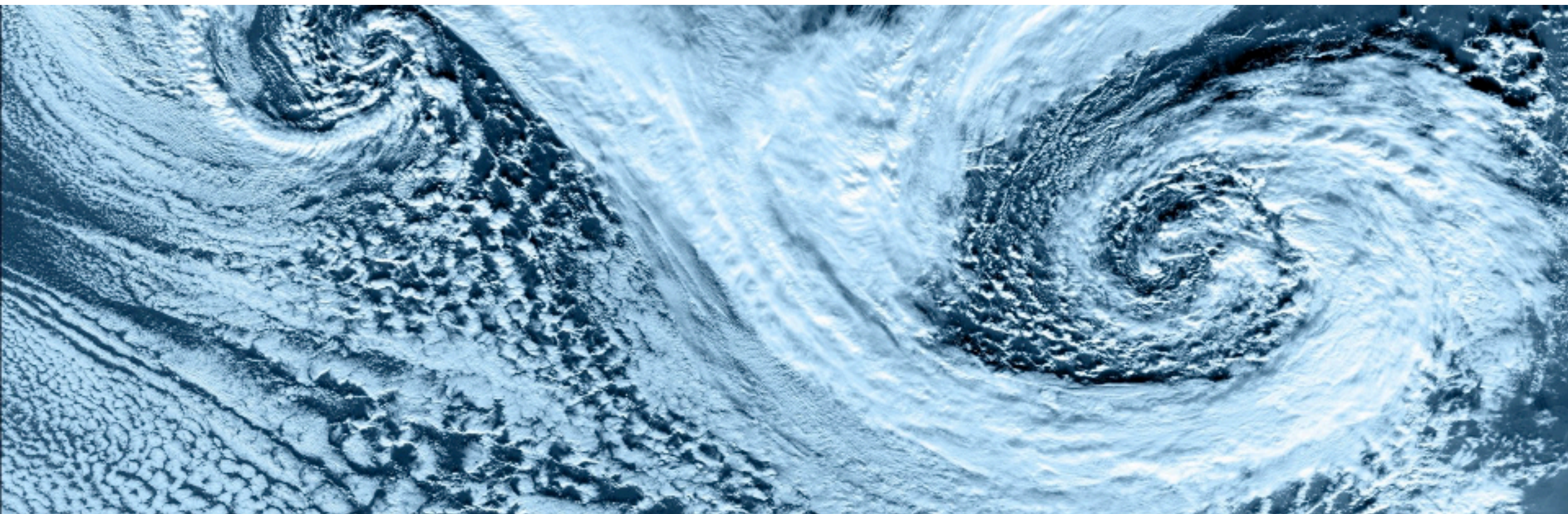


Git: The Basics

C2SM Git Courses · 26 March 2026

Annika Lauber, Mikael Stellio, Michael Jähn



Schedule

- **9:30 - 10:30** Introduction to Git
- **10:30 - 11:00** Git Branches
- **11:00 - 11:15** Coffee Break
- **11:15 - 12:00** Merge, Restores, Delete, .gitignore
- **12:00 - 13:00** Lunch
- **13:00 - 13:30** Remote Repositories
- **13:30 - 14:00** Merge Conflicts
- **14:00 - 15:00** Repository Managers

Part 1:

Introduction to Git

Why Use a Version Control System?

- Tracks file changes in a documented way
 - Restore previous versions
 - Understand what happened / changed between different versions
 - Record reasons for changes
- Allows to work simultaneously on the same code (when using remote server)
 - Alone (on different computers)
 - Multiple people in a collaboration
- Maintain several parallel versions of the same code in a systematic way
- Many tools available (web -based services, graphical interfaces, etc.)

Different Version Control System (VCS)

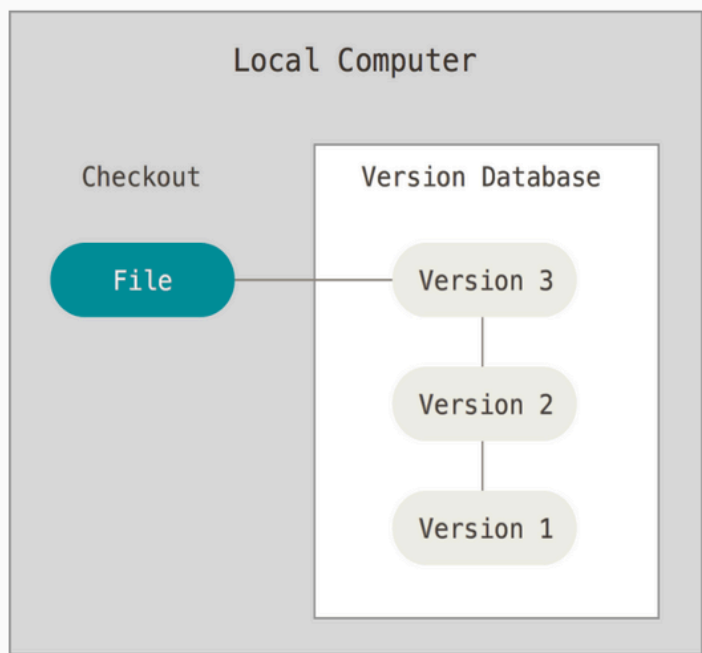
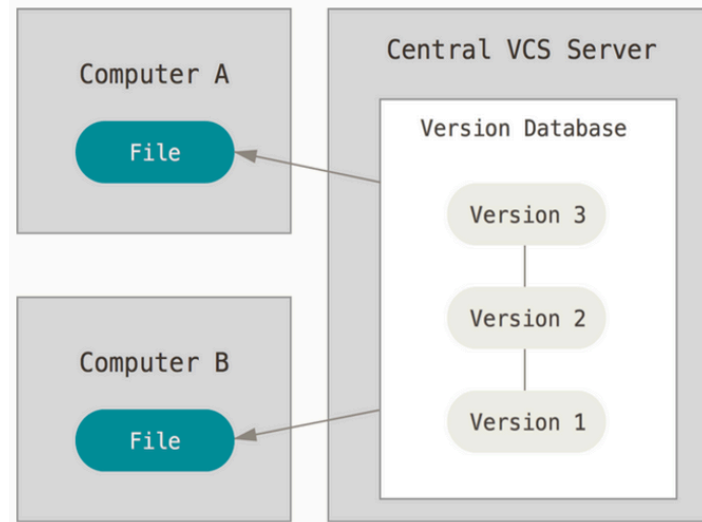
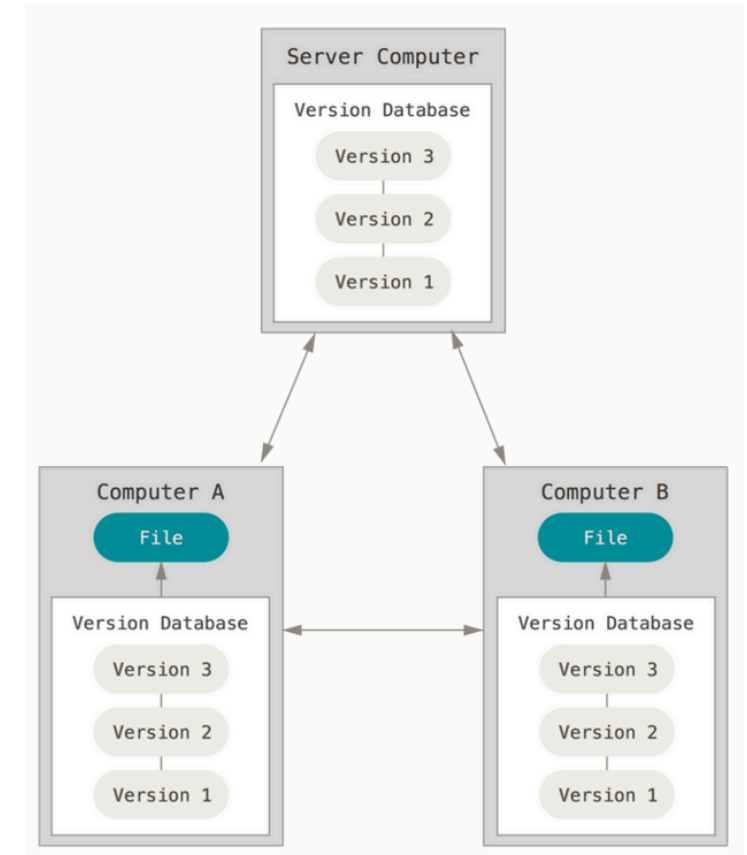


Figure 1. Local version control.

Local VCS



Centralized VCS (e.g., SVN)

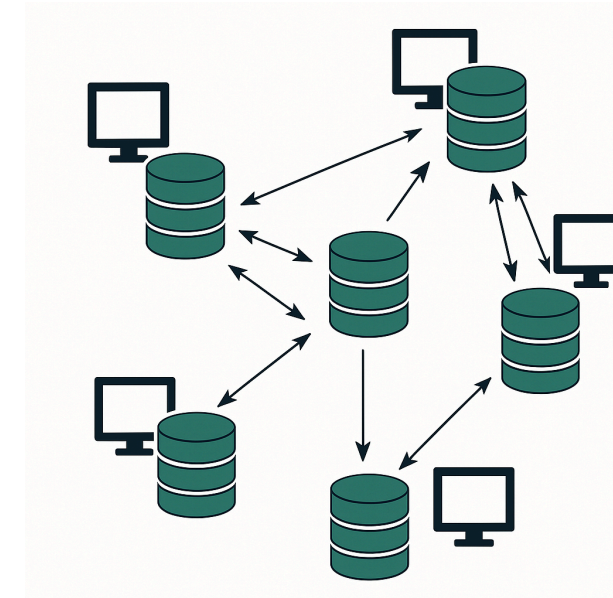


Distributed VCS (e.g., Git)

Source: book *ProGit*

Why Git?

- Designed for collaborative, open-source workflows
- Distributed
 - Easy and flexible code exchange
 - Back-up your code
- Widely used, fast and efficient



Happy Birthday, Git!

Secure, fast, flexible – and no alternative: git turns 20 years old

Linus Torvalds published git 20 years ago today. Since then, it has become ubiquitous and an integral part of software development.

"I'm an egotistical bastard and name all projects after myself. First 'Linux', now 'git'", says Linus Torvalds about himself. git is British slang for "stubborn people who think they're always right and argue around".

<https://www.heise.de/en/news/Secure-fast-flexible-and-no-alternative-git-turns-20-years-old-10340526.html> 

Practical Example: File for the Planing of a Conference

Without versioning

```
conference_schedule_v0.txt  
conference_schedule_v1.txt  
conference_schedule_v2.txt  
conference_schedule_v2_dj.txt  
conference_schedule_v2_orch.txt  
conference_schedule_v3.txt
```

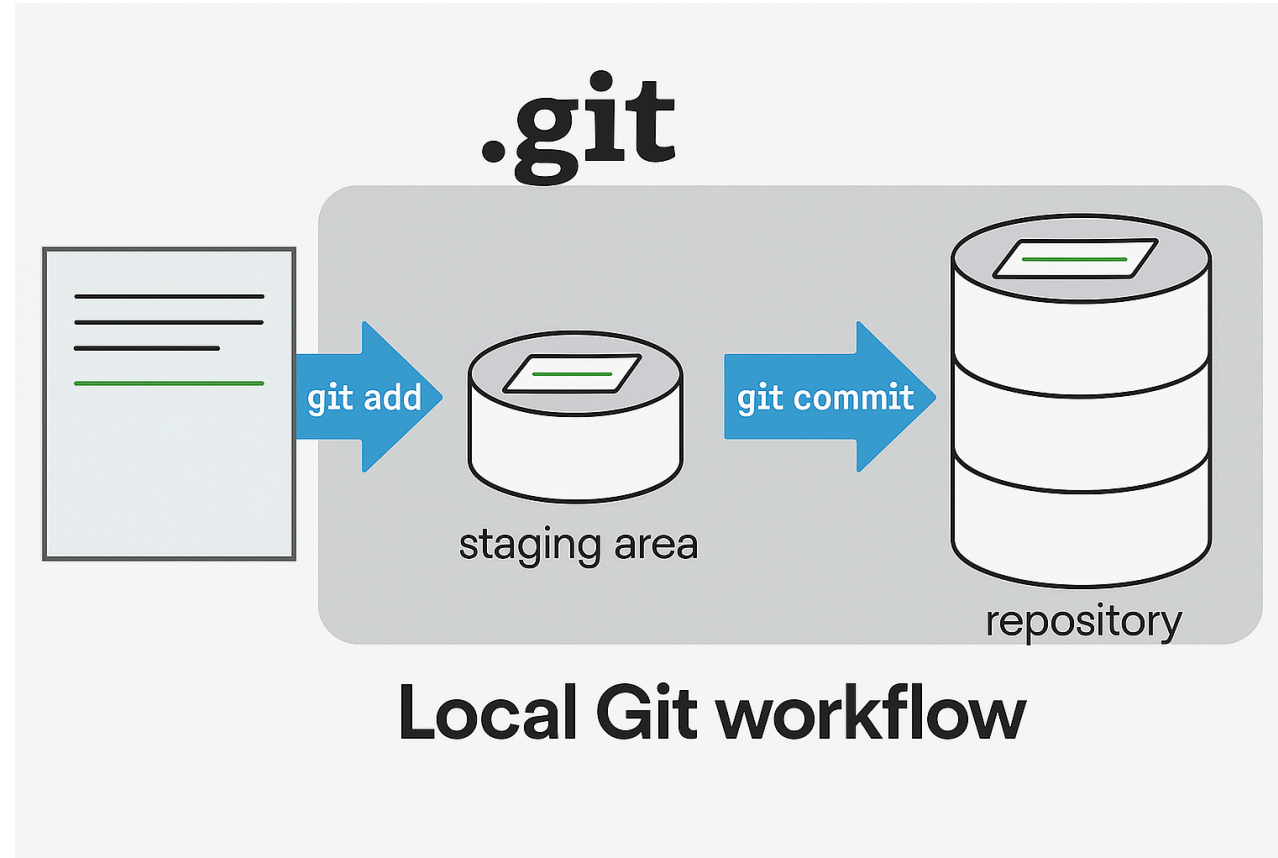
- Lots of different versions of same file
- Who remembers which one is what?

With versioning

```
conference_schedule.txt  
.git (stores version history, date and  
authors of changes, etc.)
```

- Only one working file
- History of file documented
- Easy to combine

Local Git Workflow



Important Git Commands

- `git init`:
 - Creates empty Git repository
 - Creates *main* branch by default
 - Creates the *.git* folder and contents
- `git add`:
 - Saves code changes to staging area
 - Can add all or only some of the current code changes
- `git commit`:
 - Saves changes from staging area to the repository
 - Creates a unique commit ID
 - Saves a log message for the user

Commit Messages

	COMMENT	DATE
○	CREATED MAIN LOOP & TIMING CONTROL	14 HOURS AGO
○	ENABLED CONFIG FILE PARSING	9 HOURS AGO
○	MISC BUGFIXES	5 HOURS AGO
○	CODE ADDITIONS/EDITS	4 HOURS AGO
○	MORE CODE	4 HOURS AGO
○	HERE HAVE CODE	4 HOURS AGO
○	AAAAAAAAA	3 HOURS AGO
○	ADKFJSLKDFJSDKLFJ	3 HOURS AGO
○	MY HANDS ARE TYPING WORDS	2 HOURS AGO
○	HAAAAAAAAAANDS	2 HOURS AGO

Important Git Commands

- `git log`
 - Displays the log of all commits
- `git status`
 - Shows the status of the working copy
 - States which files have been placed in the staging area
 - Shows which files have been modified but not placed in staging area
- `git diff`
 - Shows changes between two versions of the code

Important Remarks

It is **difficult to mess something up** with Git – don't be afraid of it!

Read what Git tells you, it is trying to help you!

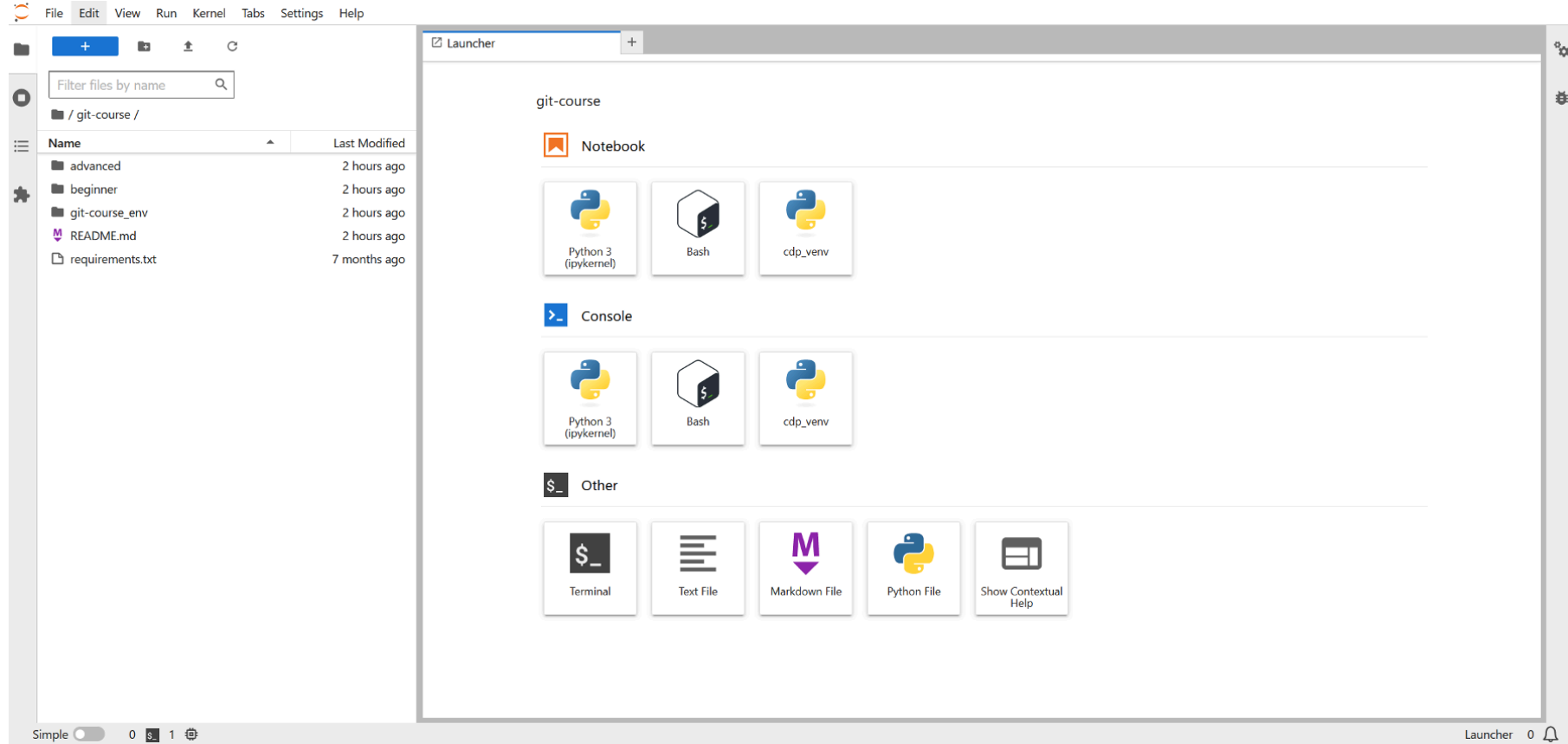
Accessing Exercises

This git course is available at:

<https://github.com/C2SM/git-course> 

- Select the «beginner» folder to access exercises,
e.g. «Exercise_1_basic_commands.md»

JupyterLab



The screenshot displays the JupyterLab web interface. On the left is a file browser pane showing the directory structure of a repository named 'git-course'. The main area on the right is the 'Launcher' pane, which provides options to create new files or notebooks. The file browser shows a search bar and a list of files and directories. The launcher is organized into three sections: 'Notebook', 'Console', and 'Other', each containing icons for different environments or file types.

File Edit View Run Kernel Tabs Settings Help

Filter files by name

/ git-course /

Name	Last Modified
advanced	2 hours ago
beginner	2 hours ago
git-course_env	2 hours ago
README.md	2 hours ago
requirements.txt	7 months ago

Launcher

git-course

Notebook

- Python 3 (ipykernel)
- Bash
- cdp_env

Console

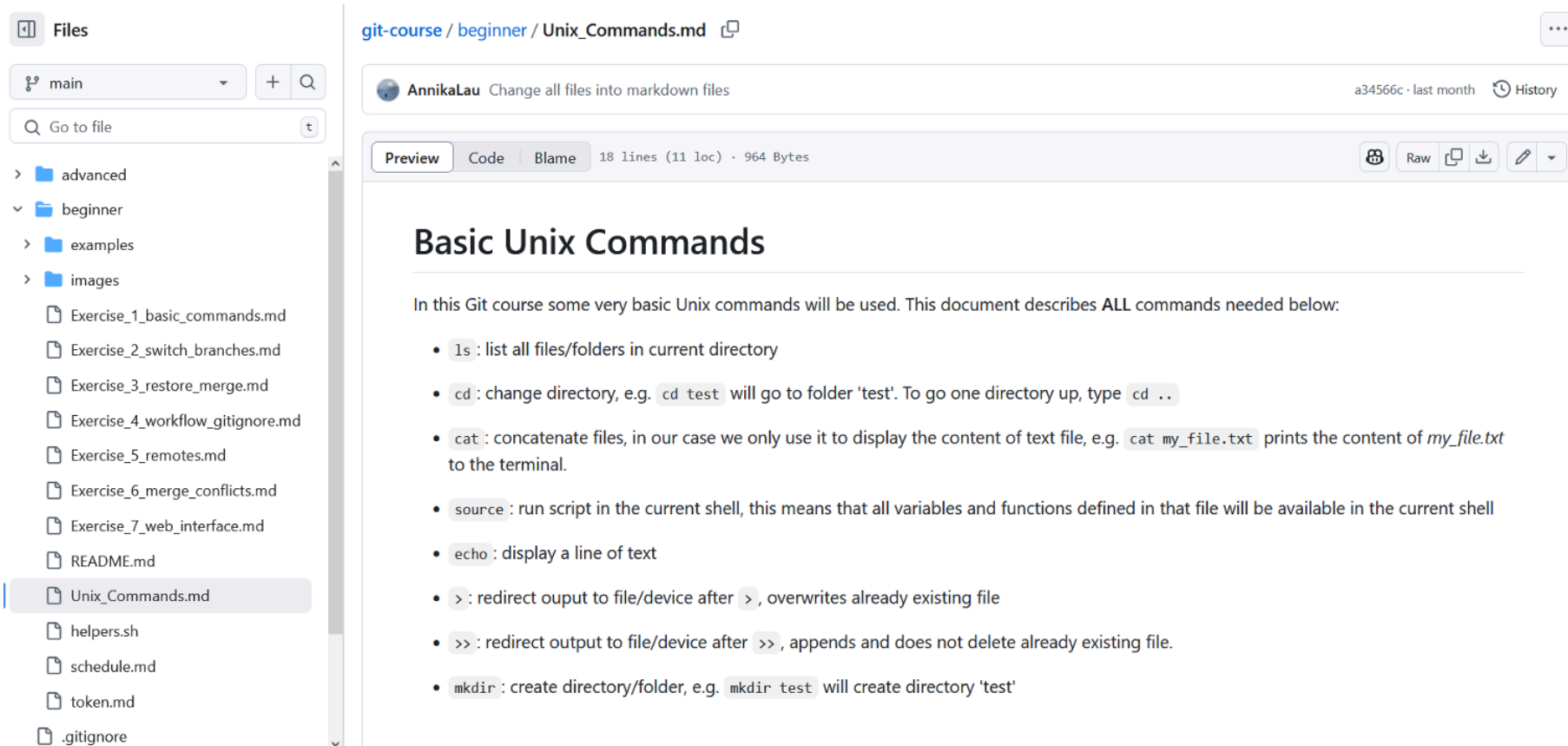
- Python 3 (ipykernel)
- Bash
- cdp_env

Other

- Terminal
- Text File
- Markdown File
- Python File
- Show Contextual Help

Simple 0 1

Launcher 0



The screenshot shows a GitHub repository interface. On the left, a file explorer shows the directory structure: `main` branch, `Go to file` search bar, and a list of files and folders including `advanced`, `beginner` (selected), `examples`, `images`, `Exercise_1_basic_commands.md`, `Exercise_2_switch_branches.md`, `Exercise_3_restore_merge.md`, `Exercise_4_workflow_gitignore.md`, `Exercise_5_remotes.md`, `Exercise_6_merge_conflicts.md`, `Exercise_7_web_interface.md`, `README.md`, `Unix_Commands.md` (selected), `helpers.sh`, `schedule.md`, `token.md`, and `.gitignore`.

The main content area shows the file `git-course / beginner / Unix_Commands.md` by user `AnnikaLau` with commit `a34566c` from last month. The file is 18 lines (11 loc) and 964 Bytes. The `Preview` tab is active, showing the title `Basic Unix Commands`.

The content of the file is as follows:

In this Git course some very basic Unix commands will be used. This document describes ALL commands needed below:

- `ls` : list all files/folders in current directory
- `cd` : change directory, e.g. `cd test` will go to folder 'test'. To go one directory up, type `cd ..`
- `cat` : concatenate files, in our case we only use it to display the content of text file, e.g. `cat my_file.txt` prints the content of *my_file.txt* to the terminal.
- `source` : run script in the current shell, this means that all variables and functions defined in that file will be available in the current shell
- `echo` : display a line of text
- `>` : redirect output to file/device after `>`, overwrites already existing file
- `>>` : redirect output to file/device after `>>`, appends and does not delete already existing file.
- `mkdir` : create directory/folder, e.g. `mkdir test` will create directory 'test'

Exercise 1

- Create Git repository from scratch
- Track changes in files using `git add`, `git status` and `git commit`
- Know state of Git repo using `git diff` and `git log`

Part 2:

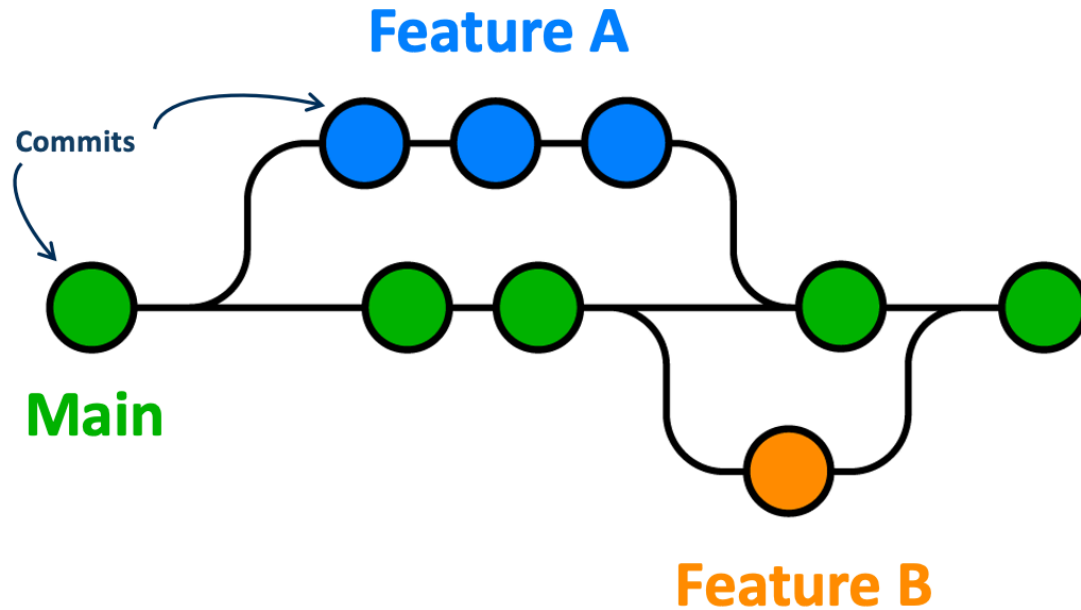
Git Branches

Remark

- Git $\geq$ v2.23: `git switch` and `git restore`.
- Older versions: `git checkout` for both.

We will use `git switch` and `git restore` for simplicity
(`git checkout` commands in brackets)

Git Branches

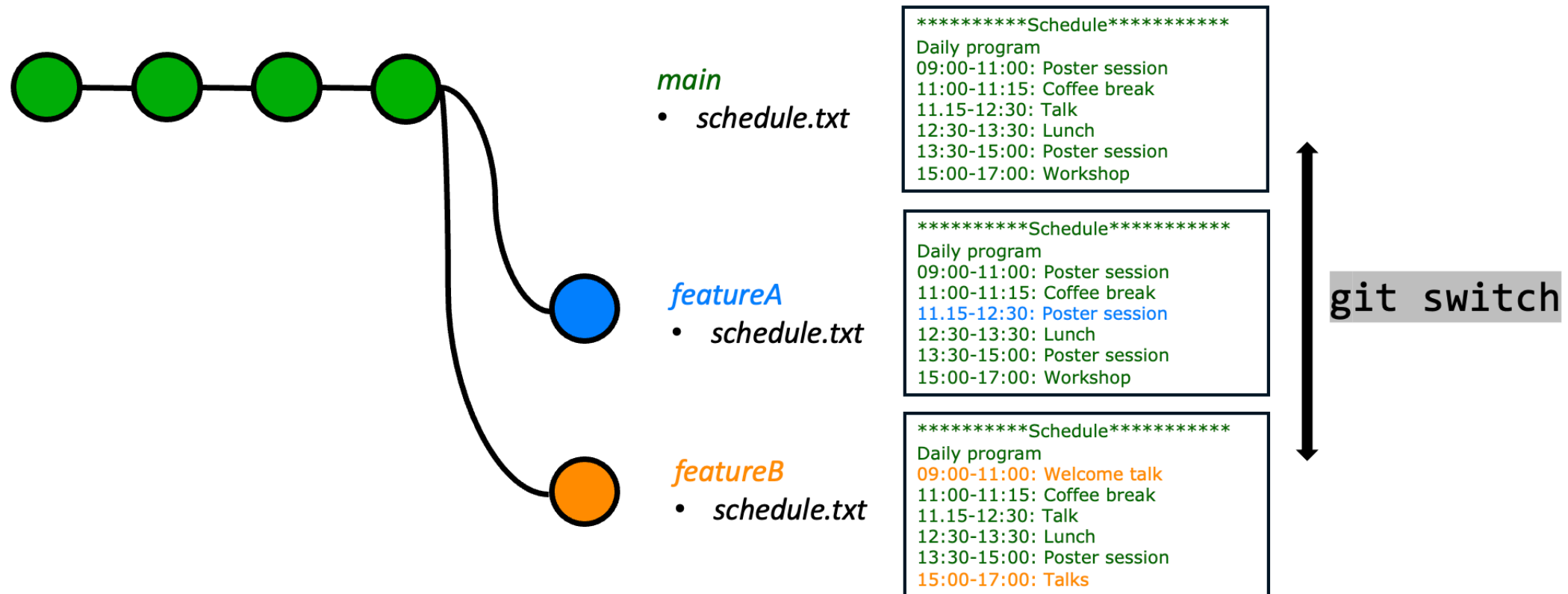


Adapted from <https://blog.nobledesktop.com/learn/git/git-branches> 

- Multiple versions of your code
- Branch = copy of your code
- Git: create, merge and delete branches
- Advantages:
 - Easy collaboration with others
 - Experiment with new features
 - Revert changes if necessary

Git Branches

- Make changes, without affecting the main version
- Multiple branches co-exist in parallel (different versions of same file)



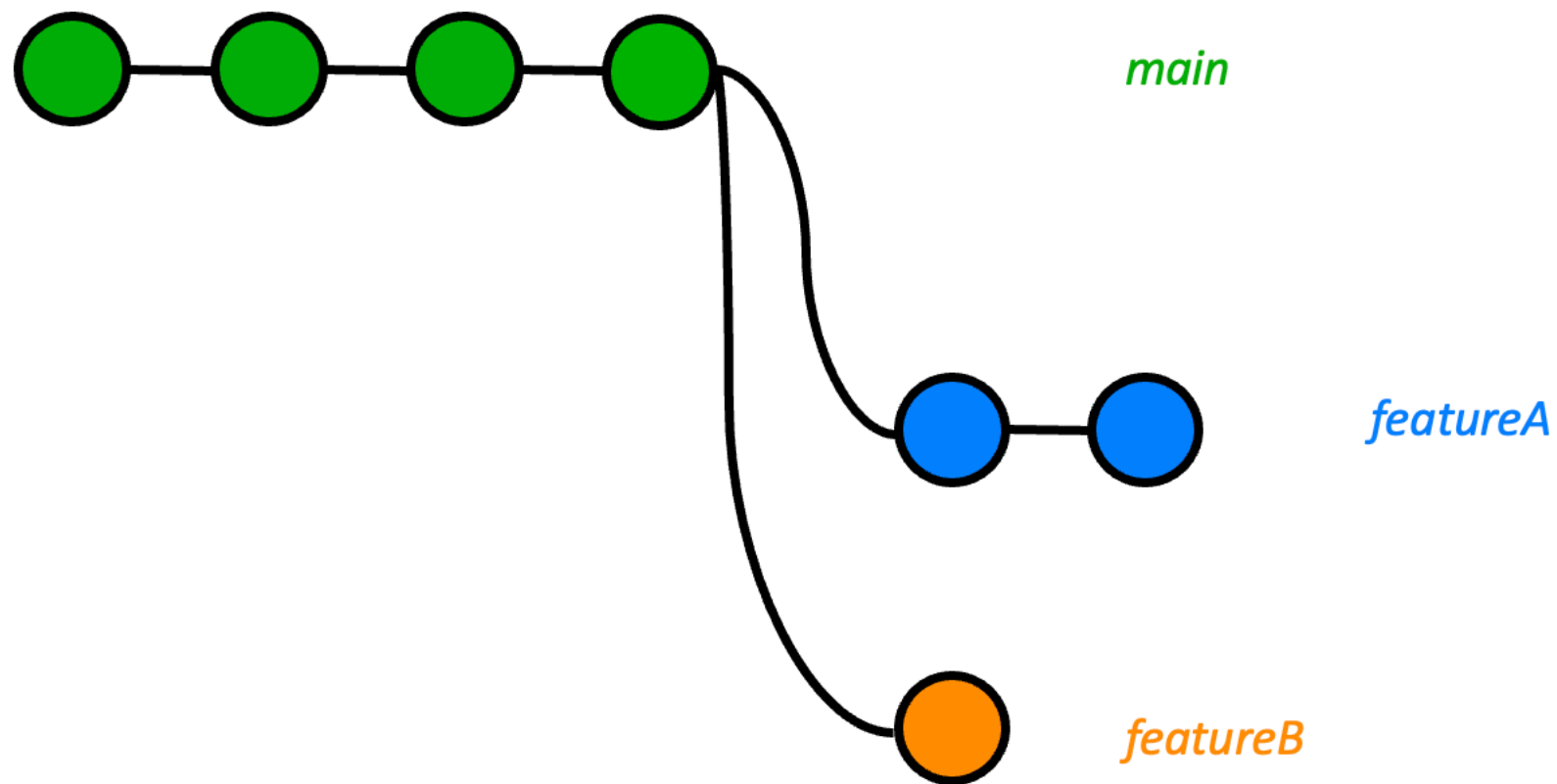
Exercise 2

- Work with branches
- Switch between branches using `git switch`

Part 3:

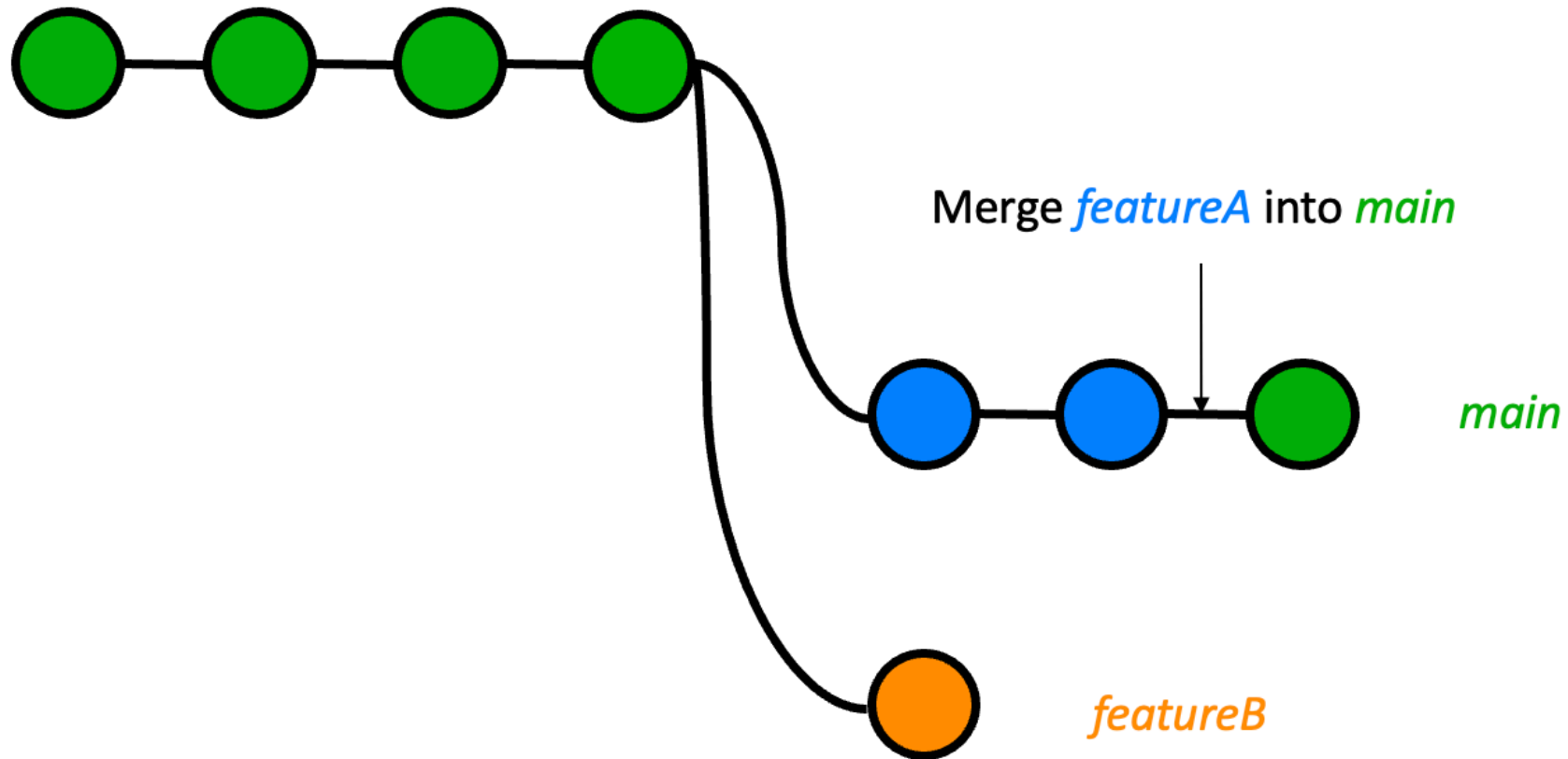
Merge, Restore and Delete

Merge Branches



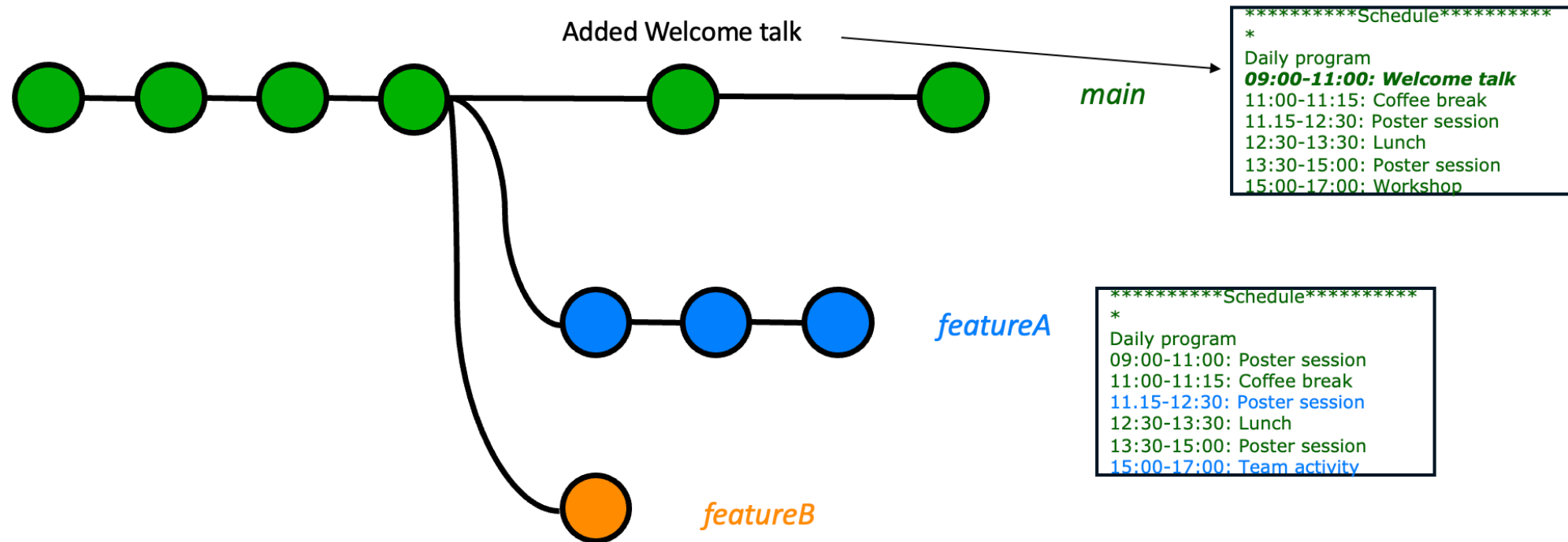
Fast-Forward Merge

- Linear history between *main* and *featureA*



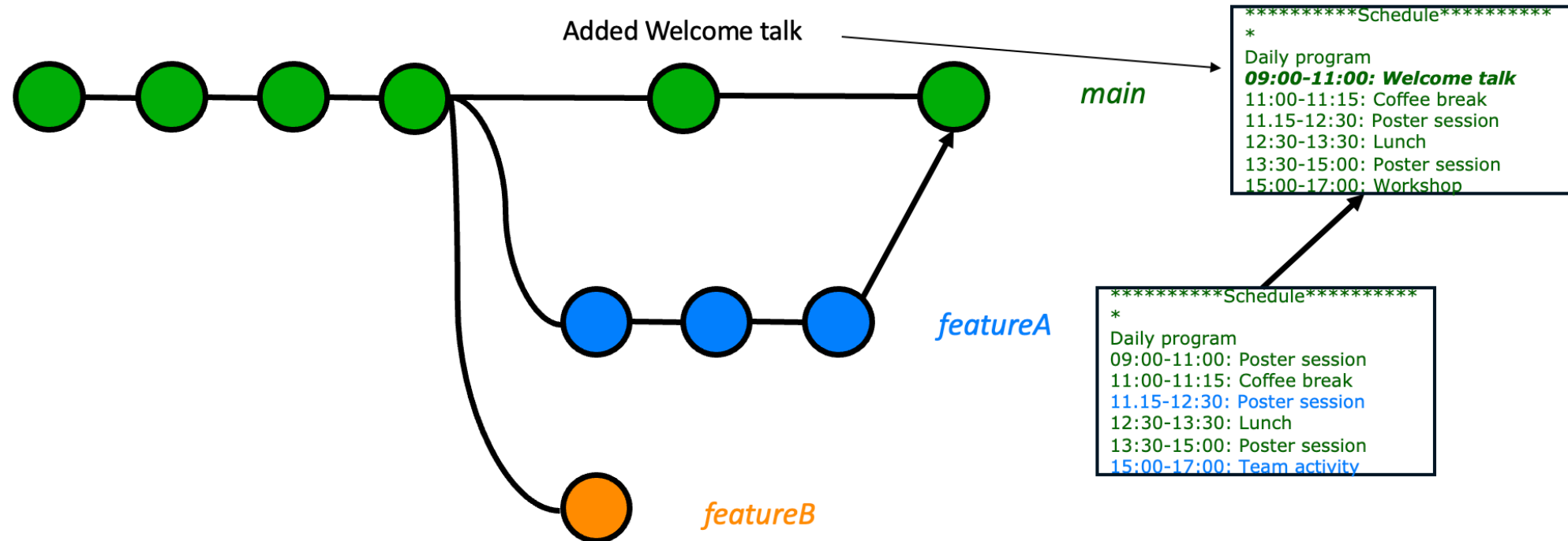
3-Way Merge

- Commits between *main* and *featureA*



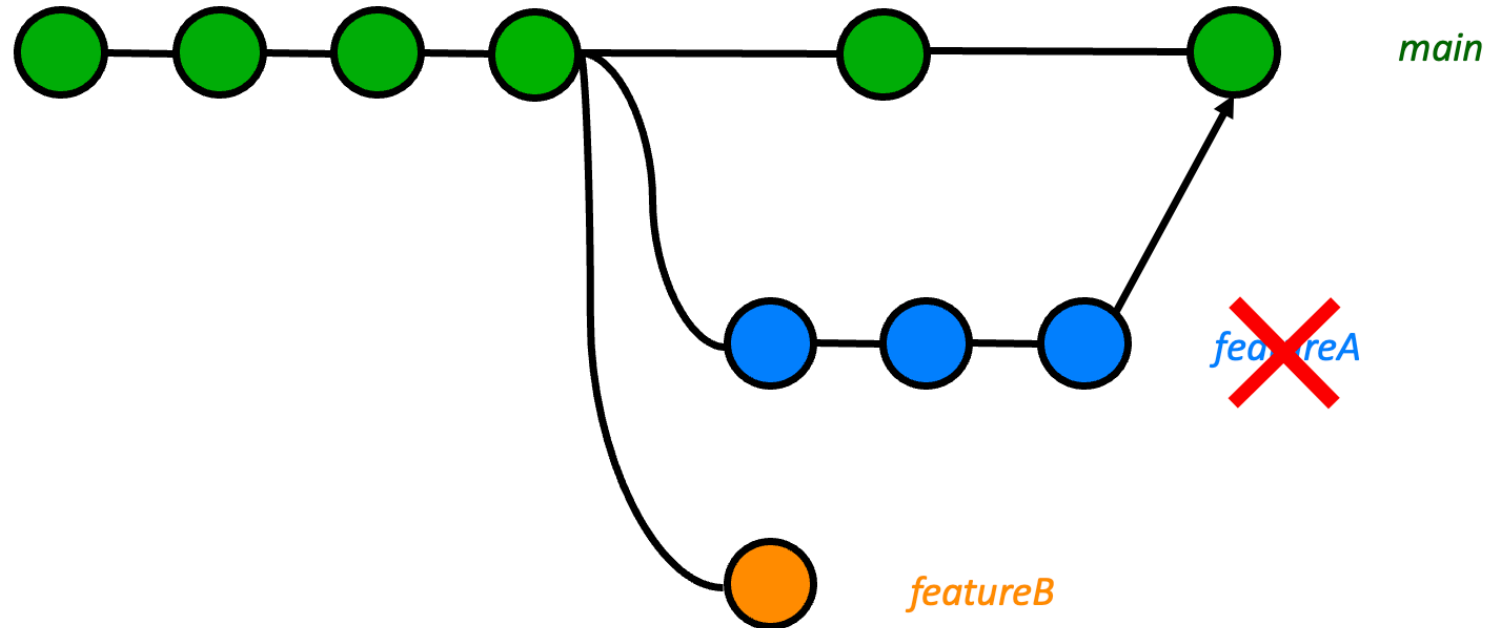
3-Way Merge

- Commits between *main* and *featureA*

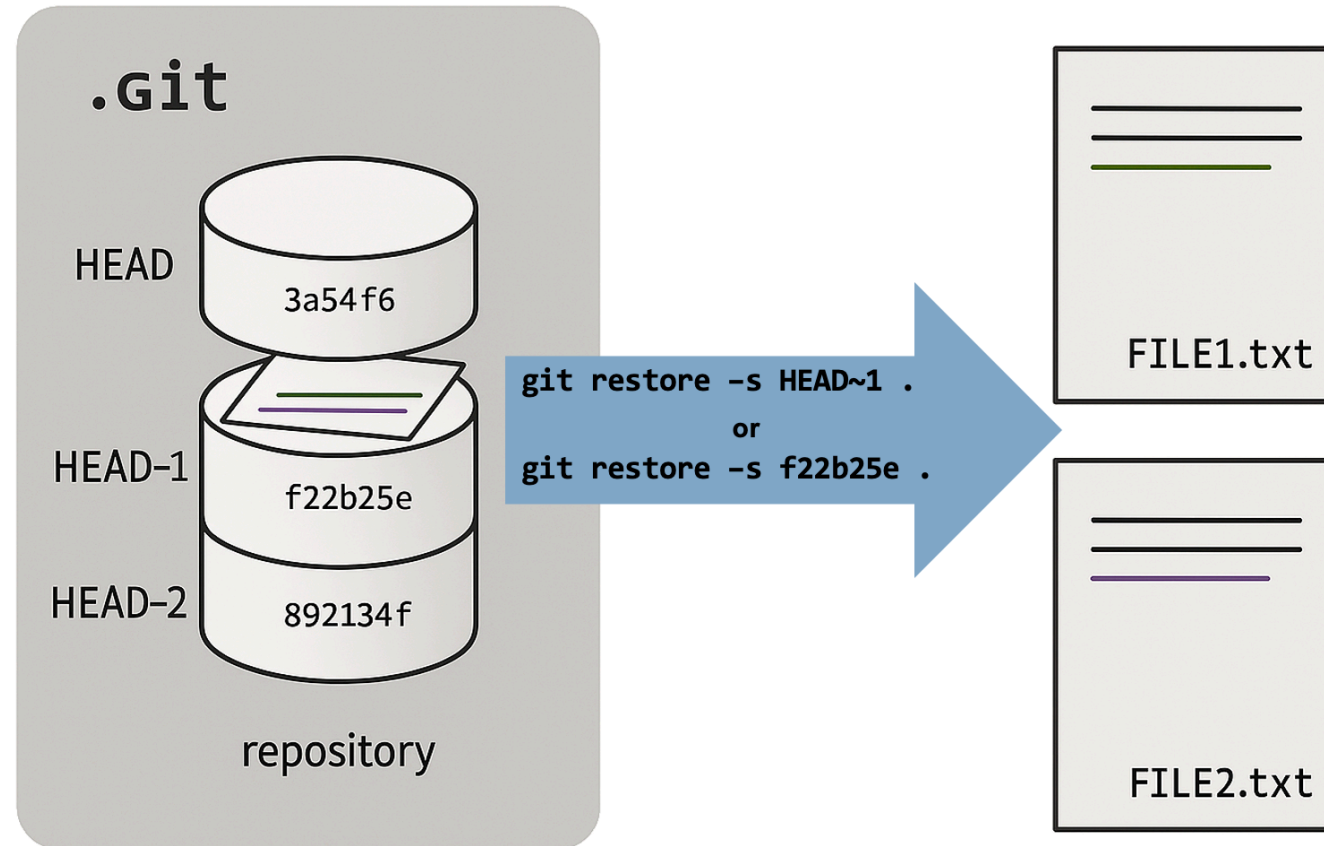


3-Way Merge

- Delete branch after merge



Git Restore



Important Git Commands

- `git restore`
 - Deletes unstaged changes in working directory
 - `-s <commit ID>` : Changes file to the state at specific commit ID across branches or back in history
 - Can be used for single files (`<file_x>`) or the entire repository (`.`)
- `git merge <source branch>`
 - Combines source branch with target branch (= current branch)
 - Does not create new commit unless specified or needed (depends on your config settings)
 - Called from target branch
- `git branch`
 - list branches in local repository
 - `-a` : list all branches (remotes included)
 - `-d` : delete branch (if already merged into HEAD)
 - `-D` : delete branch (force)

Part 4:

Practice workflow and .gitignore

Understanding .gitignore

- Specifies intentionally untracked files by Git
- Helps maintaining a clean repository
- Examples: binary files, temporary files, log files, etc.
- Example .gitignore:

```
*.log      # ignore all log files  
.DS_Store  # ignore macOS system files  
temp/      # ignore ALL files in ANY directory named temp
```

Important Git Commands

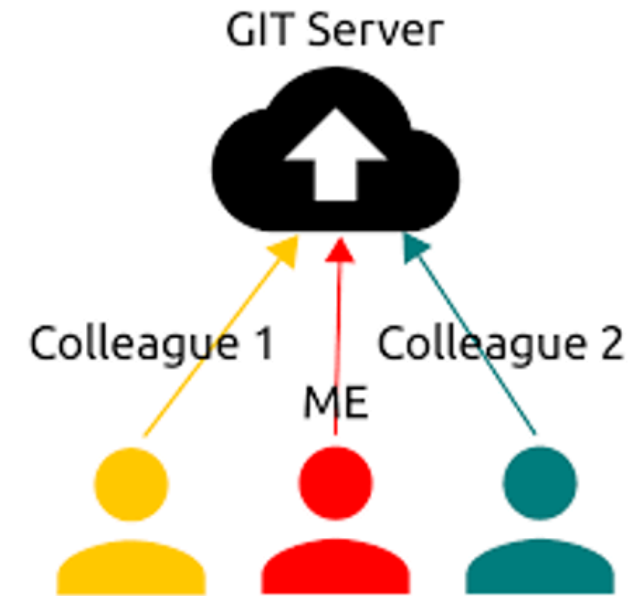
- `git init` : creates a new repository
- `git status` : show the status of working tree (gives list of modified files)
- `git diff` : show differences between commits
- `git add` : add file to the selection of files to be committed (staging area)
- `git commit` : action to create a commit (snapshot of your project at a certain time)
- `git log` : show history of the commits
- `git switch` : switch branches
- `git restore` : restore working tree files
- `git merge` : merge changes from another branch

Part 5:

Remote Repositories

Remote Repositories

- Git repositories are usually hosted on a server (with or without web interface)
- Users clone repository to work on it locally
- The Git server is a "remote" for the local repository

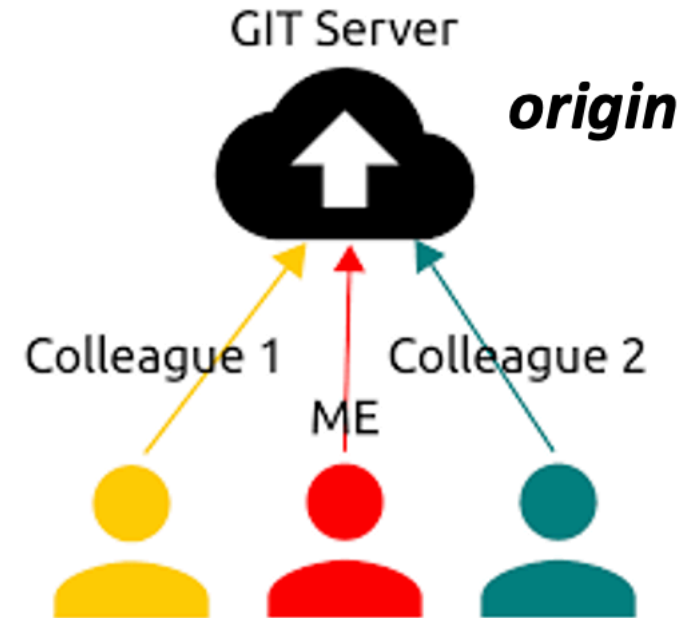


Source: <https://medium.com/swlh/continuous-integration-with-arduino-ce3714c19b44>



Remote Repositories

- Git automatically connects remote and local repositories when cloning
- The remote repository is called "origin" by Git

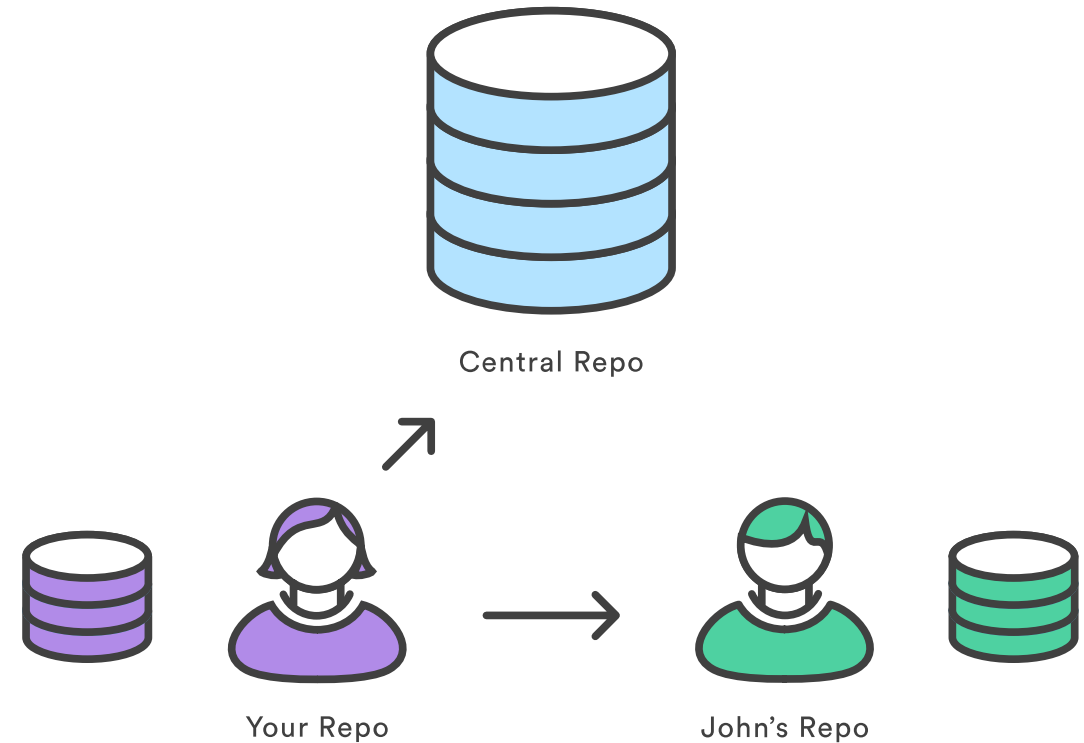


Source: <https://medium.com/swlh/continuous-integration-with-arduino-ce3714c19b44>



Remote Repositories

- Remotes don't have to be web-hosted
- Can have multiple remote repositories



Source: <https://east.fm/refcards/git/sync/syncing.html> 

Remote Branches

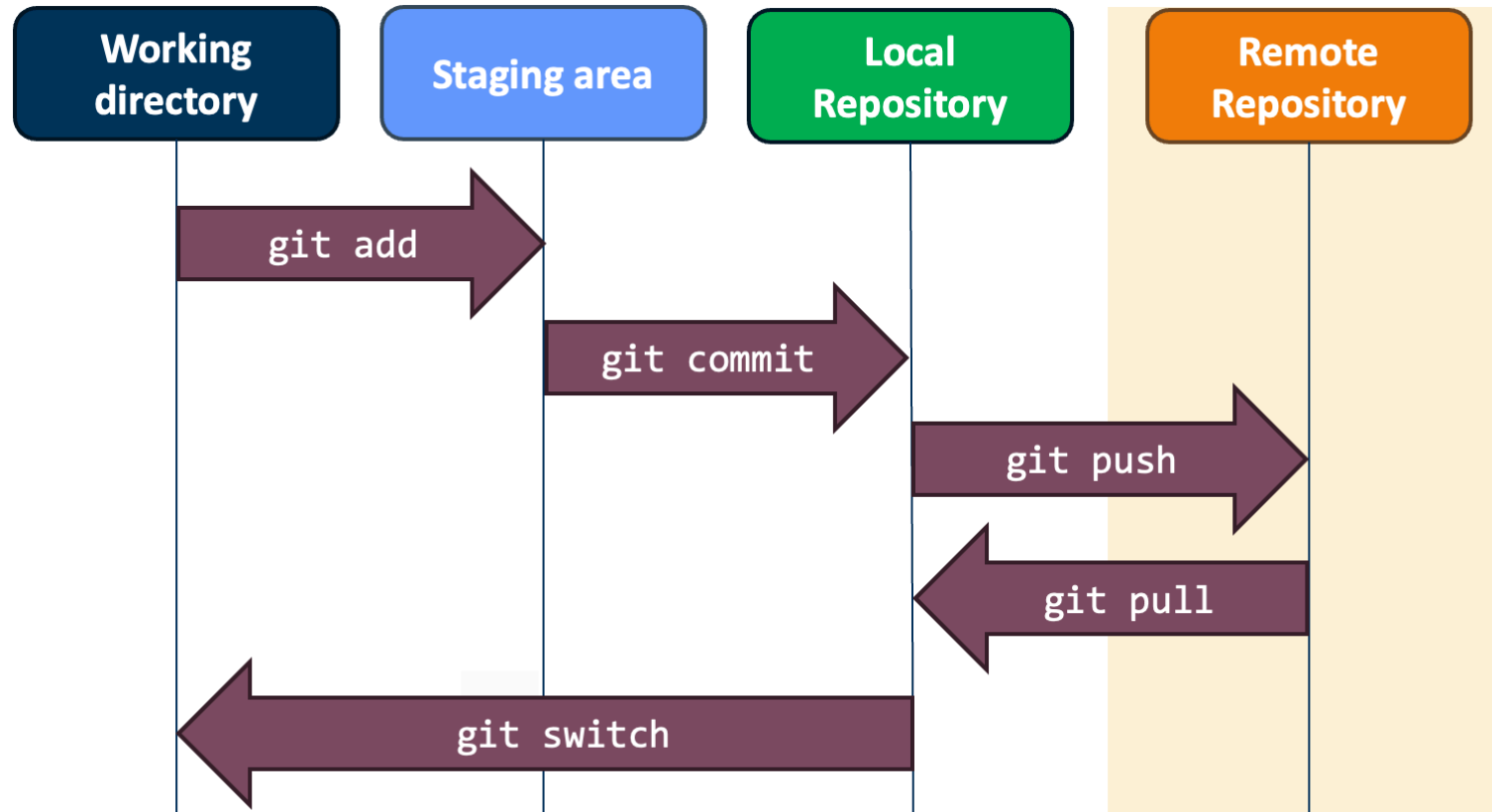
- Your local repository will have local branches AND remote branches
- You can set up local branches to track the remote ones

```
# use "git branch -a" to view ALL of the branches in your local repository
```

```
git branch -a
```

```
* main  
remotes/my_remote/main  
remotes/my_remote/updated_schedules
```

Exchanging Code with Remote Repositories



`git pull`
=
`git fetch`
+
`git merge`

Modified after <https://salesforcecodex.com/salesforce/useful-git-commands/> 

Exercise 5: Remote Repositories

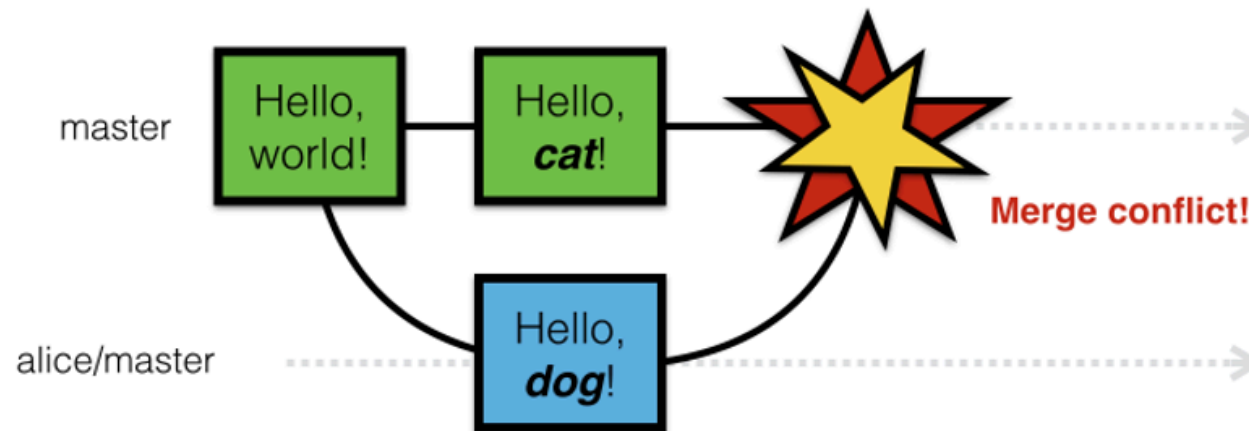
- Add a remote repository
- Examine remote branches
- Exchange information with a remote repository

Part 6:

Merge conflicts

Merge conflicts

- Modifications have been made on both branches you are merging
- AND
- If you have both modified the same line in a file or one person has deleted a file another has modified



Merge conflicts

- Git will stop the merge and notify you that you have conflicts
- You must resolve these conflicts before the merge can be completed.

```
Auto-merging Hello.f90
CONFLICT (content): Merge conflict in Hello.f90
Automatic merge failed; fix conflicts and then commit the result.
silvertk@lavoisier:~/Documents/Git_tutorials/example> git status
On branch master
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)

        both modified:   Hello.f90

no changes added to commit (use "git add" and/or "git commit -a")
```

Merge conflicts

- To resolve conflicts, there are basically two options:

1. Choose the local or the remote version of a file:

```
git restore my_file --ours/theirs
```

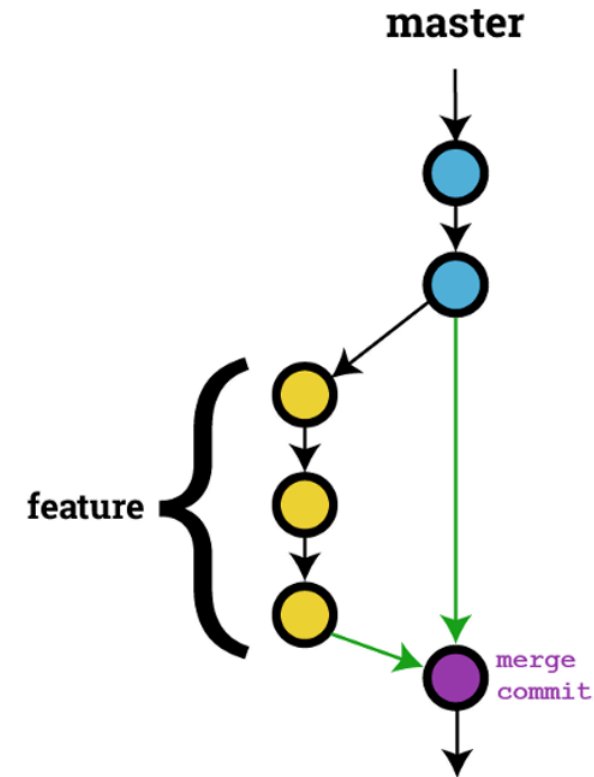
```
(git checkout my_file --ours/theirs)
```

2. Open file, find conflict markers and decide how to solve the conflict:

```
<<<<<<< HEAD  
Hello Cats!  
=====  
Hello Dogs!  
>>>>>>> branch
```

Merge Conflicts

- Use `git add` to indicate that the file is ready for the merge
- Once all the conflicted files are resolved and added to the staging area, you use `git commit` to finalize the merge



<https://haydar-ai.medium.com/learning-how-to-git-merging-branches-and-resolving-conflict-61652834d4b0> 

Exercise 6: Merge Conflicts

- Learn how to deal with merge conflicts:
 - Undo merge
 - Choose preferred version
 - Adapt file directly

Part 7:

Repository Managers

Repository Managers

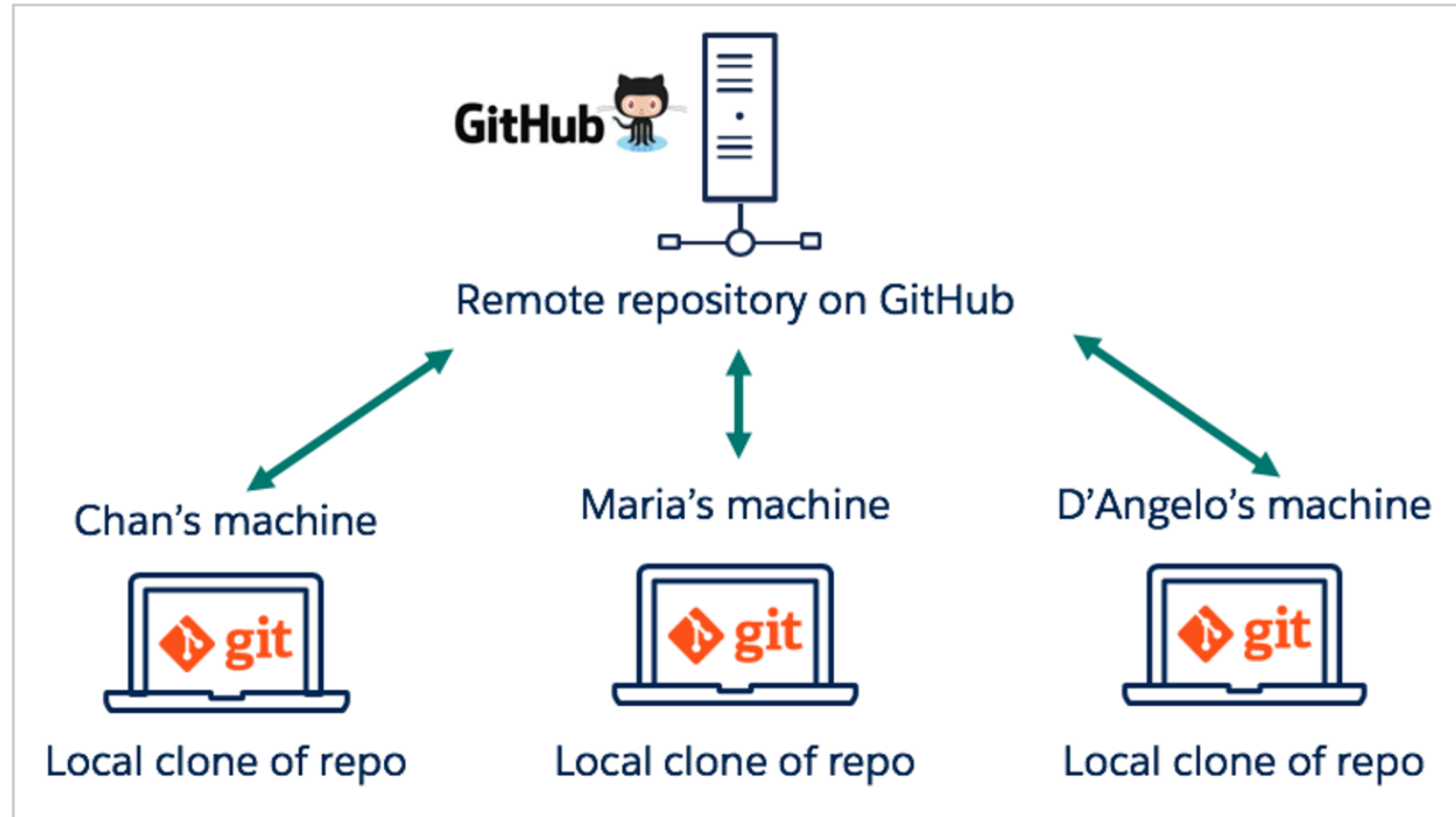
- Interfaces for Git repositories
- Can use the web service or install locally
- Pricing varies – check for academic discounts!
- Examples:
 - C2SM hosts software on GitHub web interface
 - Empa and IAC use local Gitlab servers



Repository Manager Features

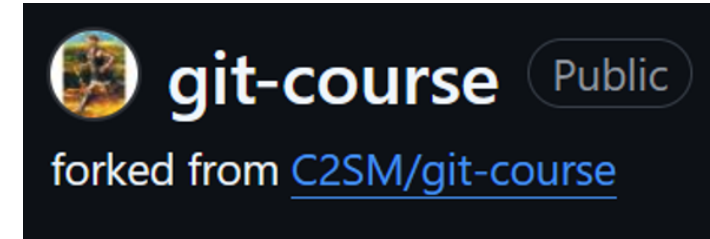
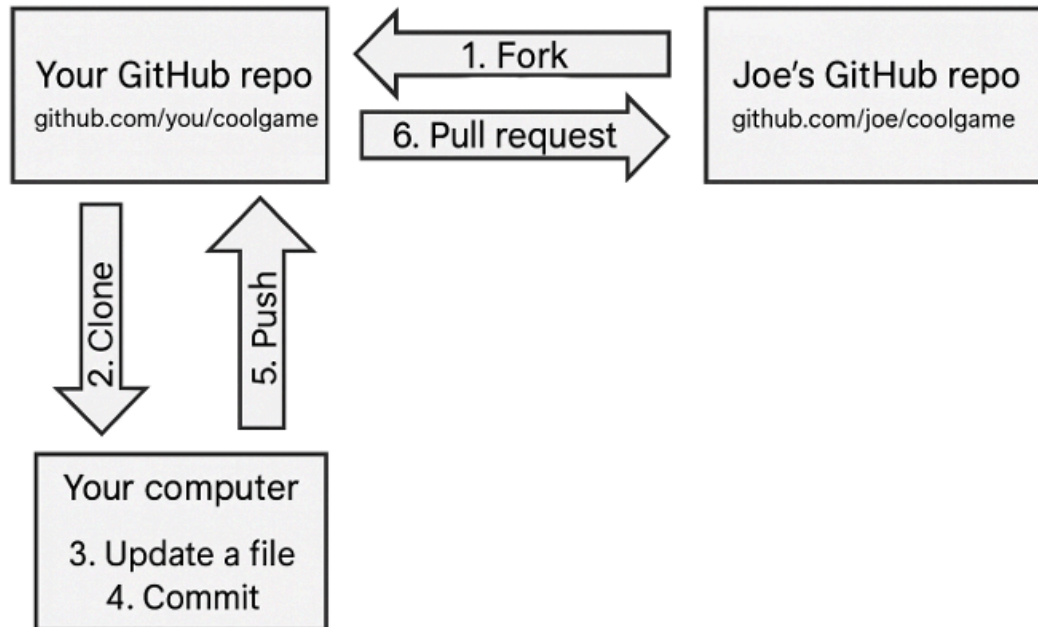
- Easily examine commits and files
- Collaboration tools (more later!)
- Issue reporting
- Code review
- Pull/merge requests
- Automation of testing

Single User or Small Group Workflow



<https://blog.devgenius.io/entering-into-devops-01-git-basics-2898fbbb4b5c> @

Large Group Workflow: Using Forks



<http://www.programmersought.com/article/4701192164/> @

Exercise 7: Repository Managers

- Access code from a Git web interface
- Push code changes to a Git web interface
- Examine the code repository on a Git web interface